

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1897

No Maries 69 kV
December 31, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1897. The Interconnection Customer proposes a 48.7 MW solar photovoltaic facility interconnecting at the No Maries 69 kV substation in the North Carolina area. The request was received on October 05, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 622 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,368,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Trowbridge Substation 138-69 kV Transformer Bank T1	Bel Air / Fabrica Tenere 138kV Line	141.0	92	25.9
Jajome 3403 Substation 138 kV/69 kV Autotransformer Bank 1	Jajome 3403 138/69 kV Transformer Bank Unit 2	138.0	145	61.6
Trowbridge 138/69 kV TX	Fabrica Tenere / Bel Air 138kV	135.8	96	25.9
Jajome 3403 138/69 kV Auto	Jajome 3403 138/69kV Auto	122.1	164	61.6
No Maries / Trowbridge 138kV	Bel Air / Fabrica Tenere 138kV	120.7	235	51.8
Fabrica Tenere Substation / Bel Air 138kV	No Maries / Trowbridge 138kV	117.3	249	50.4
City West Diesel Plant / Roberts Sw Sta 345kV Line	Roberts Sw Sta / Nauvoo 345kV	107.0	1,471	6.2
Home / No Maries 138kV	Bel Air / Fabrica Tenere 138kV	106.1	300	10.4
S Meadow / Monroe Ne 345kV	Monroe Ne / Pocket N 345kV	105.1	1,187	6.2
Victory Way / Wheelabrator Portsmouth 138kV line	Wheelabrator Portsmouth / Gosport 138kV Line	101.9	306	7.3
Bel Air / Home 138kV	No Maries / Trowbridge 138kV L	101.9	258	50.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Trowbridge 138-69 kV Transformer Bank	Replace transformer with higher-rated unit	\$10,741,000	10% (\$1,095,000)
Jajome 3403 Substation 138 kV/69 kV TX T1	Replace transformer with higher-rated unit	\$6,565,000	9% (\$609,000)
Trowbridge 138/69 kV TX	Replace transformer with higher-rated unit	\$5,747,000	10% (\$586,000)
Jajome 3403 138/69 kV Auto	Replace transformer with higher-rated unit	\$5,806,000	9% (\$538,000)
No Maries / Trowbridge 138kV	Rebuild line with higher-capacity conductor	\$15,855,000	10% (\$1,616,000)
Fabrica Tenere Substation / Bel Air 138kV	Rebuild line with higher-capacity conductor	\$19,477,000	9% (\$1,733,000)
City West Diesel Plant / Roberts Sw Sta 345kV Line	Reconductor line	\$3,922,000	0% (\$19,000)
Home / No Maries 138kV	Reconductor line	\$3,873,000	2% (\$84,000)
S Meadow / Monroe Ne 345kV	Reconductor line	\$5,292,000	0% (\$29,000)
Wheelabrator Portsmouth / Victory Way 138kV	Reconductor line	\$2,972,000	2% (\$67,000)
Bel Air / Home 138kV	Reconductor line	\$10,110,000	10% (\$992,000)
	Total allocated to project		\$7,368,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.